

Flux Logic Computing

Concept Note — Version 2.0

Deterministic Magnetic Flux Patterns as a Digital Logic Family

Gate Family Addendum — Continuation of V1.0

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Status: Conceptual — architecture level, device-agnostic

Disclaimer

This document presents a conceptual logic architecture for technical evaluation. It does not claim experimental validation, commercial readiness, or superiority over established semiconductor logic families. Performance, security, energy, and manufacturability advantages described here should be treated as hypotheses requiring simulation, measurement, and peer review.

1. Purpose and Scope

This document extends **Flux Logic Computing v1.0** by defining a concrete flux-domain AND gate and showing how a complete Boolean gate family can be derived from that primitive. The scope is intentionally architectural: it specifies signal encoding, threshold behavior, regeneration requirements, composability rules, and validation criteria while remaining independent of any single device technology.

Excerpt from V1.0: "Flux Logic Computing proposes a conceptual digital logic family where binary information is represented, propagated, and processed using deterministic magnetic flux patterns instead of voltage ranges. This concept focuses on the logic-family abstraction layer, not the physical device layer."

The goal of V2.0 is to make the gate-level proposal more precise: first by specifying a threshold-based AND construction, then by deriving NAND, OR, NOR, XOR, XNOR, buffer, fanout, and a 1-bit full adder using flux-domain operations and regenerative interfaces.

2. Notation, Encoding and Primitives

2.1 Signal encoding

- **Flux unit:** F (abstract unit).
- **Logical 0:** $\Phi_0 = F$.
- **Logical 1:** $\Phi_1 = 2F$.

2.2 Superposition

Input fluxes superpose linearly in a shared interaction region:

$$\Phi_{\text{tot}} = \sum \Phi_{\text{in}}$$

2.3 Threshold device

A threshold flux Φ_T is a nonlinear response point such that:

- If $\Phi_{\text{tot}} \geq \Phi_T \rightarrow$ device outputs Φ_1 .
- If $\Phi_{\text{tot}} < \Phi_T \rightarrow$ device outputs Φ_0 .

Design rule example. Choose Φ_T so that $3F < \Phi_T < 4F$ (e.g., $\Phi_T = 3.5F$). This ensures only the combined input of two logic-1s crosses the threshold in the AND construction below.

2.4 Primitive elements

- **Input coils / emitters** — produce canonical flux patterns Φ_0 or Φ_1 .
- **Shared core / interaction region** — where input fluxes superpose.
- **Sense / threshold element** — detects Φ_{tot} and triggers regeneration.
- **Output generator (regenerator)** — writes canonical Φ_0 or Φ_1 into the next stage.

3. Assumptions, Feasibility Boundaries and Validation Targets

The constructions in this note rely on several engineering assumptions that must be validated before the architecture can be considered physically viable. These assumptions are not presented as solved implementation details; they define the minimum technical conditions required for meaningful simulation and prototyping.

- **Canonical flux states must be separable.** The physical implementation must produce two repeatable states corresponding to Φ_0 and Φ_1 with sufficient separation after process variation, temperature drift, load variation, and noise.
- **Superposition must be predictable.** The interaction region must support controlled summation of input contributions without uncontrolled saturation, hysteresis, or cross-coupling that changes the intended truth table.
- **Threshold response must be sharp and repeatable.** The sense element must switch reliably within the selected threshold window and maintain guard bands across operating conditions.
- **Regeneration is mandatory.** Each gate output must be restored to a canonical flux pattern before driving subsequent stages; otherwise accumulated analog distortion would undermine digital composability.
- **Fanout and loading must be bounded.** Output generators must be specified with maximum load, coupling efficiency, and regeneration margin so downstream stages do not disturb the source state.

Initial validation should therefore measure or simulate: truth-table correctness, threshold margin, switching delay, regeneration fidelity, fanout limit, energy per operation, thermal stability, external magnetic leakage, and sensitivity to fabrication tolerances.

4. Flux AND Gate — Formal Definition and Physical Sketch

3.1 Logical definition (flux mapping)

Inputs $A, B \in \{0, 1\}$ are encoded as Φ_A, Φ_B . The output $Y = A \wedge B$ is encoded as Φ_Y .

Table 1 — AND gate truth table (flux values)

A	B	Φ_A	Φ_B	Φ_{tot}	Output
0	0	F	F	2F	Φ_0 (0)
0	1	F	2F	3F	Φ_0 (0)
1	0	2F	F	3F	Φ_0 (0)

A	B	Φ_A	Φ_B	Φ_{tot}	Output
1	1	2F	2F	4F	Φ_1 (1)

Choose Φ_T with $3F < \Phi_T < 4F$ (e.g., 3.5F). Only the 1-1 case (4F) crosses the threshold and produces Φ_1 .

3.2 Conceptual physical sketch

- Two input coils (A and B) are wound on, or coupled to, a shared core or tightly coupled micro-cores.
- Input coils produce canonical flux patterns Φ_0 or Φ_1 ; their fields superpose in the interaction region.
- A sense coil monitors Φ_{tot} . A nonlinear core response (saturable core, domain flip, or superconducting switching) is tuned to Φ_T .
- If $\Phi_{tot} \geq \Phi_T$, the sense stage triggers the output generator to write canonical Φ_1 into the output loop; otherwise the output generator writes canonical Φ_0 .
- The sense $\rightarrow$ generate path regenerates canonical patterns to preserve composability and noise margins.

3.3 Threshold response (concept inset)

Response vs. Φ_{tot} exhibits a marked threshold at $\Phi_T = 3.5F$: below threshold the device remains in the 0 output state; above threshold it switches to the 1 output state.

5. Deriving a Complete Gate Set from AND

A complete Boolean family can be derived from the AND primitive when paired with a flux-domain inverter. The following derivations separate logical completeness from implementation difficulty: NAND-based composition provides a conservative route, while threshold-tuned variants may reduce gate count if their physical margins can be maintained.

4.1 NOT (inverter)

Method A — Bias + threshold inversion (less robust)

Add a fixed bias flux Φ_b into the sense region so that the combined flux with the input produces the opposite decision at the chosen threshold. This requires careful bias tuning and is sensitive to drift.

Method B — Differential comparator (preferred)

Use two complementary cores or a gradiometric comparator that compares input flux against a reference flux or mirrored reference. The comparator outputs Φ_1 when the input is Φ_0 , and Φ_0 when the input is Φ_1 . This is implemented by routing input and reference fluxes into a comparator sense region and using the comparator's output generator to produce the inverted canonical pattern. The differential approach is more robust to common-mode noise and drift.

4.2 NAND

NAND = NOT(AND). Build the AND gate as in §3, then feed its output into the flux inverter (Method B). NAND is functionally complete; a flux NAND alone can implement any Boolean function.

4.3 OR

Approach 1 — De Morgan via NAND

$\text{OR}(A, B) = \text{NAND}(\text{NOT}(A), \text{NOT}(B))$. Implement NOT on each input, then NAND them.

Approach 2 — Threshold variant (direct)

Use a core with a lower threshold $\Phi_{\text{T}}^{\text{OR}}$ such that $2F < \Phi_{\text{T}}^{\text{OR}} < 3F$ (e.g., $2.5F$). Then:

- $0-0 \rightarrow 2F < \Phi_{\text{T}}^{\text{OR}} \rightarrow \text{output } 0$
- $0-1 \text{ or } 1-0 \rightarrow 3F > \Phi_{\text{T}}^{\text{OR}} \rightarrow \text{output } 1$
- $1-1 \rightarrow 4F > \Phi_{\text{T}}^{\text{OR}} \rightarrow \text{output } 1$

This directly implements OR by changing the threshold of the sense element.

4.4 NOR

$\text{NOR} = \text{NOT}(\text{OR})$. Implement OR (either method), then invert.

4.5 XOR

Composition method (recommended)

$\text{XOR}(A, B) = (A \text{ OR } B) \text{ AND NOT}(A \text{ AND } B)$. Implement OR and AND as above, invert the AND result, then AND with OR.

Direct bandpass detector (advanced)

Implement a sense region that outputs 1 only when $\Phi_{\text{tot}} \approx 3F$ (i.e., exactly one input high). This requires a narrow window detector (bandpass threshold) that is more complex but can be implemented with cascaded thresholds and gating.

4.6 XNOR

$\text{XNOR} = \text{NOT}(\text{XOR})$. Invert the XOR output.

4.7 BUFFER and FANOUT

- **Buffer** — sense input flux and regenerate canonical output flux. This is the basic sense $\rightarrow$ generate stage used after every gate to restore pattern fidelity.
- **Fanout** — the output generator drives multiple output coils; ensure each load does not pull the output core below regeneration margin. Use local buffers if needed.

6. Gate Composition Rules, Noise Margins and Design Guidelines

5.1 Flux margins and thresholds

Define a flux margin M such that $\Phi_1 - \Phi_0$ is substantially larger than expected noise, drift, and process variation. Thresholds should be placed with explicit guard bands, not merely at midpoint values, so that no valid input combination falls near an ambiguous switching region.

Table 2 — Example threshold windows

Gate	Threshold window	Example value
AND	$3F < \Phi_T^{\text{AND}} < 4F$	$\Phi_T = 3.5F$
OR	$2F < \Phi_T^{\text{OR}} < 3F$	$\Phi_T = 2.5F$

5.2 Cascading and composability

Every gate must output canonical patterns; otherwise distortion accumulates. Use regenerative output generators with sufficient drive to feed the next-stage coils.

5.3 Timing and synchronization

Gates may be **event-driven** (asynchronous) or **clocked** (synchronous). Clocked operation simplifies synchronization, pipelining, and security masking. Randomized phase or jitter can be used to reduce side-channel leakage.

5.4 Layout and coupling

Use **gradiometric layouts** and mirrored loops to cancel far-field dipole moments and reduce external leakage. Keep coupling tight (near-field) to maximize internal signal and minimize external detectability. Minimize long leads that act as antennas.

6.5 Engineering constraints to quantify

- **Switching energy:** estimate energy per transition for the sense element and regenerator separately.
- **Latency:** characterize threshold detection time, regeneration time, and any required reset interval.
- **Thermal and material stability:** quantify how temperature, aging, and magnetic history affect Φ_0 , Φ_1 , and Φ_T .
- **Crosstalk:** evaluate unintended coupling between neighboring gates, interconnect loops, and shared return paths.
- **Scalability:** determine whether routing density, fanout, and leakage cancellation remain practical beyond small demonstration circuits.

7. Example Gate Network: 1-Bit Full Adder (Flux Domain)

6.1 Equations

$$\text{Sum} = \text{XOR}(A, B) \text{ XOR } C_{\text{in}}$$

$$C_{\text{out}} = \text{Majority}(A, B, C_{\text{in}}) = (A \text{ AND } B) \text{ OR } (B \text{ AND } C_{\text{in}}) \text{ OR } (A \text{ AND } C_{\text{in}})$$

6.2 Implementation outline

- Compute pairwise ANDs: $A \wedge B$, $B \wedge C_{in}$, $A \wedge C_{in}$.
- OR the pairwise products to produce C_{out} .
- Compute Sum via the XOR chain: first XOR(A , B), then XOR(result, C_{in}).
- Regenerate outputs as canonical flux patterns for downstream stages.

6.3 Block diagram (textual)

- Inputs $\rightarrow$ AND bank (pairwise) $\rightarrow$ OR $\rightarrow C_{out}$.
- Inputs $\rightarrow$ XOR chain $\rightarrow$ Sum.
- Each intermediate output passes through a regenerator (sense $\rightarrow$ generate) to restore canonical patterns.

8. Security, Stealth and Energy Harvesting Notes

- **Stealth** — gradiometric routing and mirrored loops may reduce far-field signatures, but leakage suppression must be demonstrated quantitatively against realistic sensor noise floors, distances, shielding conditions, and operating frequencies.
- **Energy harvesting** — local harvesting from induced EMF should be treated as an optional auxiliary mechanism, not a primary power assumption, until the available harvested energy is shown to exceed sensing, regeneration, and control overheads.
- **Side-channel mitigation** — symmetric routing, randomized timing, dummy flux operations, and clock jitter may reduce observable leakage; each mitigation must be evaluated against measurable magnetic, timing, and power signatures.

9. Success Criteria and Next Steps (Practical Roadmap)

To move V2.0 from concept to engineering exploration, the next phase should convert the abstract flux model into a parameterized simulation and measurement plan:

- **Define canonical patterns** Φ_0 and Φ_1 in a candidate technology, including target magnitude, tolerance, retention behavior, and read/write mechanism.
- **Design and simulate** the AND core with material parameters, coupling coefficients, threshold model, noise sources, temperature range, and load conditions.
- **Implement the inverter/comparator primitive** and demonstrate NAND.
- **Build OR/XOR via NAND composition** in simulation.
- **Evaluate** truth-table correctness, noise margins, energy per operation, delay, fanout, reset behavior, and external field leakage.
- **Prototype a 1-bit adder** in simulation or on a testbed.

10. Appendix — Compact Derivations

- **NAND:** $\text{NAND}(A, B) = \text{NOT}(\text{AND}(A, B))$.

- **OR:** $OR(A, B) = NAND(NOT(A), NOT(B))$.
- **XOR:** $XOR(A, B) = (A OR B) AND NOT(A AND B)$.
- **XNOR:** $XNOR = NOT(XOR)$.
- **NOT:** flux comparator/inverter (differential comparator preferred).
- **Buffer / Fanout:** sense and regenerate canonical flux; use buffers to drive multiple loads.

11. Figure Plan (Captions for Layout)

Concise figure captions intended to be paired with vector diagrams in a layout tool.

- **Figure 1 — Flux AND gate (detailed).** Input coils A and B, shared core / interaction region, sense coil, output generator coil. Arrows show flux contributions from A and B into the core and the Φ_{tot} direction. Inset: response vs. Φ_{tot} with threshold at $\Phi_T = 3.5F$.
- **Figure 2 — Differential comparator (flux inverter).** Input core, reference core, gradiometric comparator block, output generator. Comparator outputs Φ_1 when the input is Φ_0 and vice versa, by comparing relative flux orientation/magnitude.
- **Figure 3 — NAND (AND + inverter).** Output of the AND core block feeds the inverter block to produce NAND.
- **Figure 4 — OR (threshold variant).** Core with lower threshold Φ_T^{OR} ; 0-1 / 1-0 / 1-1 produce output 1, while 0-0 remains 0.
- **Figure 5 — XOR (two sketches).** (A) Block composition: $XOR = (A OR B) AND NOT(A AND B)$. (B) Bandpass detector: sense region tuned to $\Phi_{tot} \approx 3F$ (single-high); more complex.
- **Figure 6 — 1-bit full adder block diagram.** Pairwise AND blocks, OR block for C_{out} , XOR chain for Sum, regenerators between stages.
- **Figure 7 — Gradiometric layout for leakage cancellation.** Mirrored loops with opposite orientation cancel the far-field dipole moment and reduce external leakage.

12. Closing Notes

V2.0 is a conceptual, architecture-level extension of V1.0. It contributes a concrete threshold-based AND gate, derives a complete set of Boolean gates in the flux domain, and identifies the engineering assumptions that must be tested before the concept can advance beyond abstraction. The architecture remains deliberately device-agnostic: candidate implementations may include micro-transformers, saturable cores, spintronic elements, superconducting loops, or magneto-optic structures. The roadmap above frames the concept as a blueprint for simulation, controlled prototyping, and independent validation rather than as a completed technology.

Summary (200 Words)

Flux Logic Computing v2.0 extends Flux Logic Computing v1.0 by defining a threshold-based flux-domain AND gate and deriving a complete Boolean gate family from that primitive. Binary values are represented as canonical flux patterns ($\Phi_0 = F$, $\Phi_1 = 2F$). Input fluxes superpose in a shared interaction region, and a nonlinear threshold element regenerates canonical output states. Choosing

Φ_T between 3F and 4F yields an AND gate that outputs 1 only when both inputs are high. With a flux comparator acting as an inverter, the architecture derives NAND, OR, NOR, XOR, XNOR, buffer, fanout, and a 1-bit full adder.

The note also strengthens the engineering frame by identifying required assumptions: separable canonical states, predictable superposition, sharp thresholds, mandatory regeneration, and bounded fanout. It outlines validation targets for truth-table correctness, noise margin, switching delay, energy per operation, thermal stability, leakage, and fabrication tolerance. Security and energy-harvesting claims are presented as hypotheses requiring measurement rather than as established advantages. V2.0 remains device-agnostic and is intended as an architecture-level blueprint for simulation, prototyping, and independent experimental evaluation.